

Software Requirements Specification (SRS)

Project Title

Memory-Based Missing Person Tracking System Using Multi-Feature Adaptive Learning

1. Introduction

1.1 Purpose

The purpose of this project is to develop an intelligent computer vision system capable of locating and tracking missing persons across multiple surveillance cameras using an adaptive memory-based feature representation. Unlike conventional systems that depend primarily on facial recognition, the proposed system integrates multiple visual characteristics such as clothing appearance, gait, body shape, accessories, motion patterns, and color distributions to identify individuals even when their faces are partially or completely occluded.

The system aims to assist law enforcement agencies, airports, railway stations, shopping malls, educational institutions, hospitals, and public surveillance systems in rapidly locating missing individuals while reducing manual monitoring efforts.

1.2 Scope

The proposed system performs:

- Person detection from surveillance video
- Multi-object tracking
- Multi-camera person re-identification
- Feature extraction
- Adaptive memory creation
- Similarity matching
- Missing person search
- Real-time alert generation
- Dashboard visualization

The system will work in both real-time and offline video analysis modes.

1.3 Intended Audience

- Police Departments
- Smart City Administrations
- Security Organizations
- Research Scholars
- Computer Vision Researchers
- Educational Institutions
- Surveillance Operators

2. Problem Statement

Thousands of people go missing every year in public places including railway stations, airports, malls, hospitals, schools, and crowded urban environments. Existing surveillance systems primarily depend on facial recognition techniques to identify individuals.

However, these systems suffer from several limitations:

- Faces may be covered using masks, helmets, scarves, or sunglasses.
- Individuals frequently change their pose or viewing angle.
- Poor lighting conditions reduce facial recognition accuracy.
- Low-resolution surveillance footage makes facial identification unreliable.
- Face recognition fails when only the back or side of a person is visible.
- Appearance changes over time reduce recognition performance.
- Current tracking systems often lose identities when people move across multiple cameras.

Consequently, manual searching remains time-consuming and inefficient.

The proposed system addresses these limitations by creating a dynamic memory profile for every detected person using multiple visual features rather than relying solely on facial information.

3. Objectives

Primary Objective

To develop an adaptive memory-based computer vision system capable of accurately detecting, tracking, and re-identifying missing persons across multiple surveillance cameras without depending exclusively on facial recognition.

Secondary Objectives

- Detect people in real time.
- Track individuals continuously.
- Extract multiple visual features.
- Build adaptive memory profiles.
- Compare feature similarity.
- Perform multi-camera tracking.
- Generate real-time alerts.
- Maintain historical tracking information.
- Improve identification accuracy under occlusion.
- Reduce false matches.

4. Existing System

Current systems use:

- Face Recognition
- RFID
- GPS Tracking
- Manual CCTV Monitoring

Limitations

- Face visibility required
- High false negatives
- Poor performance under occlusion
- Cannot handle appearance variations
- Limited multi-camera tracking
- No adaptive memory mechanism

5. Proposed System

The proposed system introduces a **Memory-Based Multi-Feature Tracking Framework**.

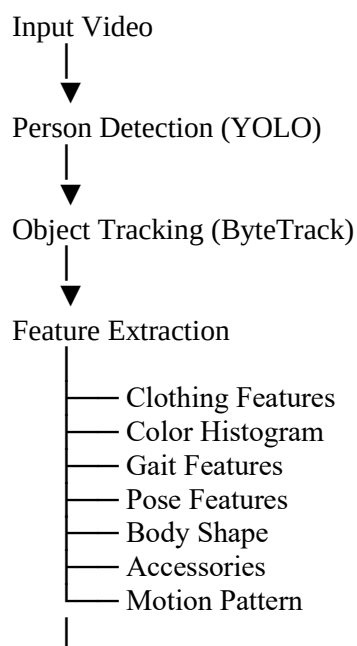
Instead of remembering only facial features, the system continuously learns multiple characteristics of an individual.

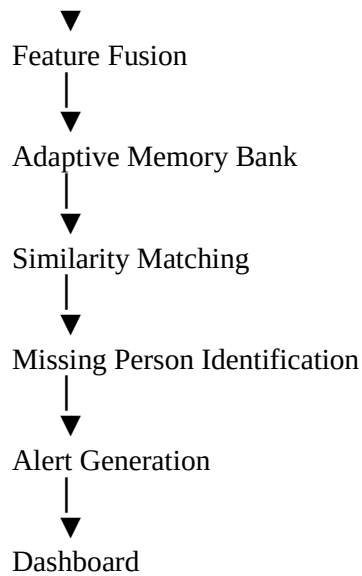
The adaptive memory contains:

- Clothing appearance
- Color histogram
- Gait pattern
- Body proportions
- Walking behavior
- Accessories
- Motion trajectory
- Historical observations
- Confidence score

The memory is continuously updated as new observations become available.

6. System Architecture





7. Functional Requirements

FR1 Person Detection

The system shall detect every person present in the surveillance video.

FR2 Person Tracking

The system shall assign a unique tracking ID to every detected person.

FR3 Feature Extraction

The system shall extract:

- Clothing features
- Body shape
- Pose
- Gait
- Color information
- Accessories
- Motion trajectory

FR4 Adaptive Memory Creation

The system shall maintain an adaptive memory profile for every tracked individual.

FR5 Multi-Camera Re-identification

The system shall identify the same person across different surveillance cameras.

FR6 Similarity Matching

The system shall compare the current observation with stored memory profiles.

FR7 Alert System

The system shall notify operators when a missing person is detected.

FR8 Search Module

Users shall search using:

- Image
- Video frame
- Person ID
- Feature similarity

FR9 Dashboard

Display:

- Live camera feeds
- Tracking IDs
- Alerts
- Detection confidence
- Last known location

FR10 History Management

Store:

- Timestamp
- Camera ID
- Person features
- Tracking history

8. Non-Functional Requirements

Performance

- Detection latency below 100 ms per frame
- Real-time processing at approximately 30 FPS
- Support multiple simultaneous cameras

Reliability

- System availability of 99%
- Robust to lighting changes and partial occlusions

Scalability

- Handle thousands of stored person profiles
- Expand to additional cameras without redesign

Security

- Role-based login

- Encrypted storage of person data
- Secure communication between modules

Usability

- Simple web dashboard
- Easy search interface
- Visual alerts and logs

9. Hardware Requirements

Minimum

- Intel Core i5
- 16 GB RAM
- NVIDIA RTX 3060 (or equivalent GPU)
- 512 GB SSD
- HD surveillance cameras

Recommended

- Intel Core i7 / AMD Ryzen 7
- 32 GB RAM
- NVIDIA RTX 4070 or better
- 1 TB SSD
- Multiple IP cameras

10. Software Requirements

Operating System

- Windows 11
- Ubuntu 22.04

Programming Language

- Python 3.11+

Libraries

- OpenCV
- PyTorch
- Ultralytics YOLO
- ByteTrack
- OpenGait
- MMPose
- FAISS
- NumPy
- Pandas
- Streamlit or Flask/FastAPI

Database

- PostgreSQL or MongoDB (metadata)
- FAISS for feature indexing

Development Tools

- Visual Studio Code
- Git
- GitHub
- Docker (optional)

11. User Requirements

Administrator

- Manage users
- Configure cameras
- View alerts
- Monitor dashboards
- Generate reports

Operator

- Search missing persons
- Monitor live feeds
- Receive alerts
- Export tracking history

12. Constraints

- Camera quality affects accuracy.
- Heavy rain, fog, or darkness may reduce performance.
- Similar clothing among different individuals can increase ambiguity.
- GPU resources are recommended for real-time inference.
- Privacy regulations must be respected when deploying in public spaces.

13. Future Enhancements

- Drone-assisted tracking
- Cross-city camera integration
- Thermal and infrared camera support
- Vision-language search (e.g., "person with a red backpack")
- Continual learning for long-term adaptation
- Mobile application for field officers
- Explainable AI showing why a match was made
- Edge AI deployment for on-camera inference

14. Expected Outcomes

- Accurate tracking without relying solely on facial recognition.
- Robust re-identification across multiple cameras.
- Faster identification of missing persons.

- Reduced manual surveillance effort.
- Improved accuracy under occlusion and appearance changes.
- A scalable framework suitable for smart city deployments and further research.

Novel Research Contribution

The key innovation of this project is an **Adaptive Multi-Feature Memory Bank** that continuously updates a person's profile by combining appearance, gait, body shape, accessories, motion patterns, and temporal history. Unlike traditional face-based systems, it dynamically adapts to appearance changes and supports robust multi-camera re-identification, making it a strong foundation for both a resume-worthy capstone project and a publishable research paper.